

Term Test 1 Version A – Solutions

A1. [2 marks] Let $P_2(x) = -2 + 5(x - 3) + 3(x - 3)^2$ be the degree-2 Taylor polynomial of $y = f(x)$ centred at $x = 3$. Find the equation of the tangent line to $y = f(x)$ at the point $(3, f(3))$.

Solution: From the Taylor polynomial, we know that $f(3) = -2$, $f'(3) = 5$. The equation of the tangent line to $f(x)$ at $x = 3$ is given by $y = f'(3)(x - 3) + f(3)$.

$$y = 5(x - 3) - 2 \text{ or you could have simplified it to } y = 5x - 17$$

A2. [2 marks] Let $f(x)$ be a function with the following properties:

$$f(4) = 3, \quad f'(4) = -1, \quad f''(4) = 6, \quad f'''(4) = -5$$

Write the degree-3 Taylor polynomial of $f(x)$ centred at $x = 4$.

Solution: The formula tells us that

$$P_3(x) = f(4) + f'(4)(x - 4) + \frac{f''(4)}{2!}(x - 4)^2 + \frac{f'''(4)}{3!}(x - 4)^3$$

$$P_3(x) = 3 - (x - 4) + \frac{6}{2}(x - 4)^2 - \frac{5}{3!}(x - 4)^3$$

$$= 3 - x + 4 + 3x^2 - 24x + 48 - \frac{5}{6}(x^3 - 12x^2 + 48x - 64)$$

$$P_3(x) = 3 - (x - 4) + \frac{6}{2}(x - 4)^2 - \frac{5}{3!}(x - 4)^3 \quad \text{or} \quad -\frac{5}{6}x^3 + 13x^2 - 65x + \frac{325}{3}$$

A3. [2 marks] A raccoon is walking along the top of a fence. Let $v(t)$ denote its velocity as a function of time (measured in minutes since midnight). Write an integral that represents the total distance travelled by the raccoon between midnight and 1 am. Your answer can contain “ $v(t)$ ” in it.

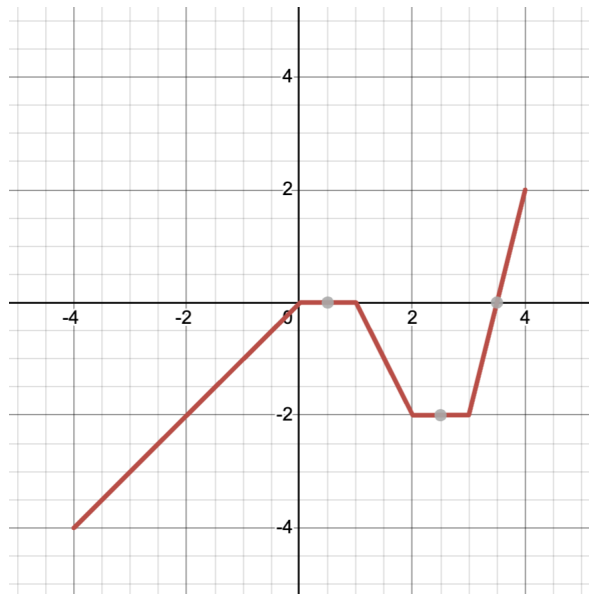
Solution: At midnight, $t = 0$ minutes. At 1 am, $t = 60$ minutes. The total distance travelled can be written as $\int_0^{60} |v(t)| dt$. Alternatively, you can let times $t = t_1, t_2, \dots, t_n$ be the times when the raccoon switches directions and write a series of integrals like this:

$$\left| \int_0^{t_1} v(t) dt \right| + \left| \int_{t_1}^{t_2} v(t) dt \right| + \dots + \left| \int_{t_n}^{60} v(t) dt \right|$$

Note that this is technically not a single integral, but a sum of multiple integrals unless the raccoon only travels in one direction.

$$\int_0^{60} |v(t)| dt$$

A4. [2 marks] Let the following graph be the graph of $y = f(x)$.



What is the average value of f over the interval $[-1, 4]$?

Solution: You can use the formula $\frac{1}{b-a} \int_a^b f(x) dx$ where $a = -1$, $b = 4$ and the integral is the net area between the graph $y = f(x)$, the x -axis, $x = -1$, and $x = 4$, which can be found by counting squares. So we get

$$\frac{1}{4 - (-1)} \int_{-1}^4 f(x) dx = \frac{(-1/2) + 0 + (-1) + (-2) + (-1/2) + (1/2)}{5} = \frac{-7/2}{5} = \frac{-7}{10} = -0.7$$

$$\frac{-7}{10}$$

A5. [2 marks] Let $g(x)$ be the rate of release of lactic acid into the bloodstream of someone as they exercise, x seconds since the exercise session started. Here, $g(x)$ is measured in mg/L. Use the following table of values to approximate the total amount of lactate added to the person's bloodstream in the first 4 seconds of exercising by writing the following Riemann sum.

Write a Riemann sum with **two (2) rectangles** created with **right endpoints**, where both subintervals have equal length.

You do not have to evaluate the sum. In other words, your answer can have “+” in it.

x	0	1	2	3	4
g(x)	85	80	90	105	100

Solution: $2(90) + 2(100)$

Also acceptable as an answer: 380

A6. [2 marks] Let $F(x)$ be an anti-derivative of $f(x) = \cos(2x + 2\pi)$. Let $F(-\pi) = \frac{-1}{2}$. Write the function $F(x)$ in the box below.

Solution: $\frac{d}{dx} \left(\frac{\sin(2x + 2\pi)}{2} - \frac{1}{2} \right) = \cos(2x + 2\pi)$ and $\frac{\sin(-2\pi + 2\pi)}{2} - \frac{1}{2} = \frac{-1}{2}$

Your answer: $\frac{\sin(2x + 2\pi)}{2} - \frac{1}{2}$

A7. [2 marks] Let $G(x) = \int_0^x e^{t^2} dt$. Write an expression for $\frac{d}{dx} \left[G(3x + 2) \right]$ as a function of x .

Solution: $\frac{d}{dx} \left[G(3x + 2) \right] = G'(3x + 2) \cdot (3) = 3e^{(3x+2)^2}$

Your answer: $3e^{(3x+2)^2}$

A8. [2 marks] Let $F'(x) = f(x)$ and let $F(0) = -7$, $F(3) = 2$, and $\int_0^3 F(x) dx = 4$.

Find $\int_0^3 x f(x) dx =$

Solution: Using integration by parts, we get $u = x$ and $dv = f(x) dx$ which implies that $du = dx$ and $v = F(x)$.

So $\int_0^3 x f(x) dx = \left[xF(x) \right]_0^3 - \int_0^3 F(x) dx = 3F(3) - 0F(0) - 4 = 3(2) - 4 = 2$

Your answer: 2

A9. [2 marks] As part of a particular weather pattern, increasing temperatures around a cloud causes it to release less snow. The rate of release of snow from 1 cubic meter of cloud per hour at temperature T is $A(T) = 3 - (2T + 1)^5$ kg snow per degree Celsius.

DO NOT EVALUATE THIS INTEGRAL:

$$\int_0^2 A(T) dT$$

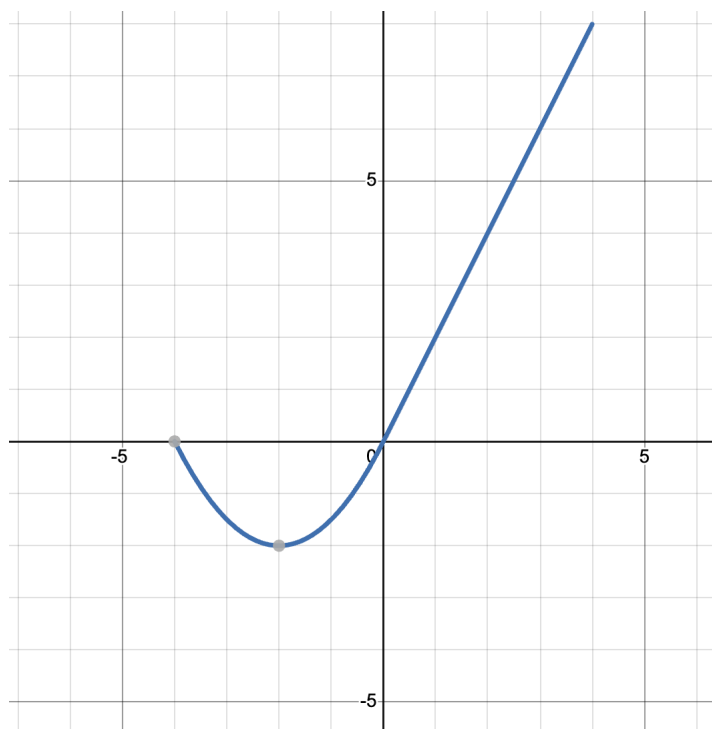
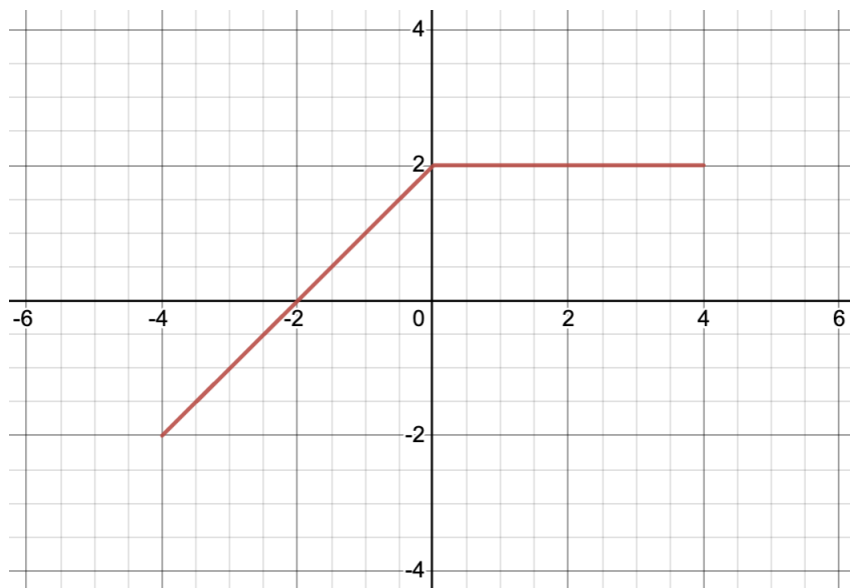
Consider the above integral. Write the unit of measurement of the integral in the box below. Use English words to describe the units.

Solution: (kg of snow from 1 cubic metre of cloud per hour at temperature T degrees Celsius) x (Number of degrees Celsius)

Your answer: **kg from 1 cubic metre of cloud per hour** at all $T \in [0, 2]$

PART B – Show your work

B1. [4 marks] Consider the following graph of the function $y = f'(x)$. On the empty axes below, sketch a possible graph for $y = f(x)$ such that $f(0) = 0$.



B2. The rate of change of the population of Wonderland t years since January 1, 2000 is modelled by the function $Q(t) = \sqrt{2t + 1}$, measured in thousands of people per year. On January 1, 2000, the population of Wonderland was 500,000.

B2.(a) [3 marks] Write an integral that is equal the total number of thousands of additional people who became part of the population from January 1, 2000 until January 1, 2024. **You will evaluate it in part (c).**

Your answer: $\int_0^{24} Q(t)dt$ or $\int_0^{24} \sqrt{2t + 1}dt$

B2.(b) [3 marks] Find all anti-derivatives of $Q(t)$.

Solution: $\frac{d}{dx} \left(\frac{1}{3}(2x + 1)^{3/2} \right) = (2x + 1)^{1/2} = \sqrt{2x + 1}$

Your answer: $\int \sqrt{2t + 1}dt = \frac{1}{3}(2x + 1)^{3/2} + c$

B2.(c) [3 marks] Evaluate the integral from part (a).

Your answer: $\int_0^{24} \sqrt{2t + 1}dt = \left[\frac{1}{3}(2x + 1)^{3/2} \right]_0^{24} = \frac{(7^3)}{3} - \frac{(1)}{3} = 114$

B2.(d) [2 marks] What was the population of Wonderland on January 1, 2024?

Your answer can be an expression that evaluates to the correct answer. You do not have to simplify it.

Solution: 614,000 or $(1000)(\text{answer from (c)}) + 500,000$

Your answer: $114,000 + 500,000 = 614,000$

B3. Consider the differential equation

$$\frac{dy}{dx} = \frac{x}{2y}$$

where $y(2) = 6$.

In this question, we will approximate a solution for y .

B3.(a) [1 mark] Calculate $y'(2)$

Your answer: $\frac{2}{2(6)} = \frac{1}{6}$

B3.(b) [2 marks] Calculate $y''(2)$

Solution: $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{x}{2y} \right) = \frac{2y - 2x \frac{dy}{dx}}{(2y)^2}$

Plug in $y = 6$, $x = 2$, and $y' = \frac{1}{6}$ your answer to part (a).
You do not have to simplify.

Your answer: $\frac{2(6) - 2(2)(\frac{1}{6})}{4(6)^2} = \frac{12 - (2/3)}{4(36)} = \frac{34/3}{144} = \frac{17}{216}$

B3.(c) [3 marks] Use your answers to parts (a) and (b) of this question to calculate a Taylor polynomial of degree 2 for the solution of the differential equation y , centered at $x = 2$.

Your answer: $6 + (\text{answer to a})(x - 2) + \frac{1}{2}(\text{answer to b})(x - 2)^2$

B3.(d) [1 mark] Use your approximation to write an expression that is equal to an estimated value of $y(2.25)$. You do not have to simplify this expression.

Solution: Substitute $x = 2.25$ into your answer to (c).
You do not have to simplify.

Your answer: $6 + (\text{answer to a})(0.25) + \frac{1}{2}(\text{answer to b})(0.25)^2$